

ANIL PRADHAN

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EDUCATION

B.Tech in Computer Science & Engineering • ITER, SOA University, Bhubaneswar CGPA: **9.11/10** | 2024 – 2028
Class XII (PCM with IT) • Odisha Board of Secondary Education **85%** | 2022 – 2024

EXPERIENCE

College Ambassador | Techfest, IIT Bombay Jun 2026 – Present

- **Represented Techfest IIT Bombay**, Asia's largest science and technology festival, by leading campus outreach and engagement initiatives to promote competitions, workshops, and technical events across student communities.
- **Conducted market research**, competitor analysis, sponsorship research, and partnership mapping, preparing professional strategy reports for corporate outreach, business development, and event collaborations.
- **Designed and developed web assets**, including landing pages, promotional content, and technical project deliverables supporting Techfest campaigns.
- **Planned and executed marketing strategies**, collaborating with technical clubs, student organizations, and online communities to increase awareness and participation.
- **Delivered data-driven recommendations** by creating presentation decks, reports, outreach strategies, and execution roadmaps using AI-powered research and productivity tools.

TECHNICAL PROJECTS

AI Career Mentor | Live Demo | Solo Full-Stack Developer March 2026 – Present

- **Multi-Agent Orchestration & Concurrency**: Engineered a high-concurrency Multi-Agent Career Platform orchestrating 7 specialized AI agents using a LangGraph Parallel Directed Acyclic Graph (DAG); reduced end-to-end execution latency by 60% (from ~4 mins to 60s) by fanning out resume parsing and live market scraper nodes concurrently.
- **System Resilience & Fault Tolerance**: Implemented a robust failover architecture featuring custom Circuit Breakers and automated API fallback chains (Groq ■ Cerebras ■ NVIDIA NIM) for downstream LLM calls, ensuring high platform availability and graceful degradation during peak-load API rate limit (429) spikes.
- **Real-time Bidirectional Streaming & State Machines**: Designed a full-duplex WebSocket messaging protocol integrating the Gemini Multimodal Live API (gemini-2.5-flash-native-audio) for low-latency voice coaching, and built a 7-Phase Finite State Machine (FSM) mock interview simulator with an integrated Monaco IDE code sandbox and real-time audio synthesis.
- **High-Efficiency RAG & Semantic Search**: Optimized an offline-first Retrieval-Augmented Generation (RAG) pipeline utilizing ChromaDB with local ONNX runtime embeddings (all-MiniLM-L6-v2) to eliminate API vector search costs, implementing an auto-switching keyword search fallback to prevent Out-Of-Memory (OOM) crashes on resource-constrained hostings.
- **Rate-Limiting, Caching & Telemetry**: Architected a secure backend middleware tier utilizing Upstash Redis for sliding-window rate limiting and active token validation, and integrated structured server diagnostics (Loguru) with a custom telemetry system tracking live API success rates, latency matrices, and token cost rollups.
- **Data Durability, Dockerization & CI/CD**: Enforced strict data integrity with Pydantic v2 validation schemas at every agent boundary, persisting records to a serverless PostgreSQL database with PgBouncer connection pooling; containerized services via multi-stage Docker builds and automated test validations (116 Pytest suite) via GitHub Actions CI/CD.

Task Pilot-AI | Live Demo | Integration Lead • Team IdeaForg-E June 2026 – July 2026

- **Solved Developer Context Overload**: Built a backend pipeline that aggregates engineering tasks from 7 different sources (Jira, GitHub, Slack, Email, Calendar, Meetings, and Incidents) into a single database, saving developers up to 2 hours daily by eliminating the need to jump between multiple tools.
- **Recovered Untracked "Hidden" Tasks**: Developed a hybrid extraction system using LLMs and local keyword rules to scan emails and Slack chats for hidden action items. This captured the ~35% of tasks that usually get lost or buried in chat histories and meeting notes.
- **Reduced LLM Costs and API Failures**: Cut down LLM token usage and resolved Groq API rate limits (429 errors) by building local fallback algorithms. This included using a local fuzzy text-matching algorithm to merge duplicate tasks and parsing Slack messages locally without hitting paid APIs.
- **Prevented Database Locking and Workflow Freezes**: Configured SQLite with Write-Ahead Logging (WAL) to ensure multiple agents could write data simultaneously without locking the database, and created a self-healing monitor that automatically kills and restarts pipeline runs stuck for over 5 minutes.
- **Automated Real-Time Emergency Re-Prioritization**: Built a chat command API that allows developers to type a new issue (e.g., "inject a P1 defect"). The backend parses the text, creates the event, and autonomously re-runs the entire 6-stage pipeline to re-rank and update calendars in under 15 seconds.

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, SQL
- **Frontend:** Next.js, React, Tailwind CSS, Axios
- **Backend:** FastAPI, LangGraph, LangChain Core, Pydantic, SQLAlchemy 2.0, HTTPX, Loguru, WebSocket, Uvicorn, Prometheus, Sentry, SlowAPI, pdfplumber, edge-tts, Postman Test Suite, Pytest
- **AI / LLM & RAG:** Groq (Llama 3.3 70B), NVIDIA NIM, Gemini Live API, Tavily, Serper.dev, DuckDuckGo
- **Databases:** Neon Postgres, SQLite, Upstash Redis, ChromaDB (Vector Store)
- **Auth & Security:** JWT (python-jose), Google OAuth 2.0, Firebase Authentication, bcrypt, CORS Middleware, Rate Limiting, HTTPS TLS, .env Secrets
- **DevOps & Infra:** Docker, GitHub Actions (CI/CD), Vercel, Render.com, GHCR
- **Tools:** VS Code, Git, Swagger UI, Cline, GitHub Copilot, Gemini Code Assist, OpenAI Codex, AntiGravity IDE

ACHIEVEMENTS & CERTIFICATIONS

- **Google Cloud Arcade Program (2024–25)**: Completed GCP challenges, earning 75 Arcade Points across hands-on cloud labs.
- **Solved 200+ problems across GFG and LeetCode**, awarded GFG Certificate and T-shirt.
- **Active participant in Developer Communities (GDG Events) and college tech events**, exploring AI systems, cloud infrastructure, and backend engineering.
- **Competitions**: Participated in multiple IIT/NIT coding contests and hackathons focused on AI systems, backend engineering, and algorithmic problem solving.
- **Hackathons**: Built production-grade MVP solutions for Microsoft AI DevDays (\$80K+ prize pools), Amazon Nova AI (\$95K+ prize pools), and Google Solution Challenge 2026, delivering deployable full-stack systems under strict constraints.